Griffith UNIVERSITY

3031ICT: Applied Data Mining

3804ICT: Data Mining

Lecture 7: Outlier Detection



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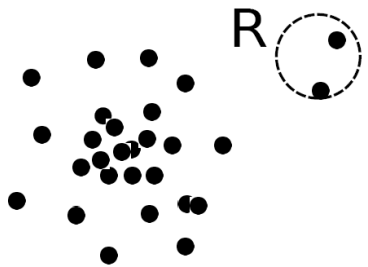
Outlier Analysis



- Outlier and Outlier Analysis
- Outlier Detection Methods
- Statistical Approaches
- Proximity-Based Approaches
- Clustering-Based Approaches
- Classification Approaches
- Mining Contextual and Collective Outliers
- Summary

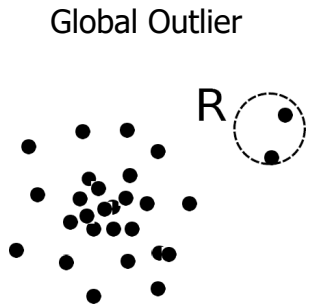
What Are Outliers?

- **Outlier**: A data object that **deviates significantly** from the normal objects as if it were **generated by a different mechanism**
  - Ex.: Unusual credit card purchase, ...
- Outliers are different from the noise data
  - Noise is random error or variance in a measured variable
  - Noise should be removed before outlier detection
- Outliers are interesting: It violates the mechanism that generates the normal data
- *Outlier detection* vs. *novelty detection*: early stage, outlier; but later merged into the model
- Applications:
  - Credit card fraud detection
  - Telecom fraud detection
  - Customer segmentation
  - Medical analysis



Types of Outliers (I)

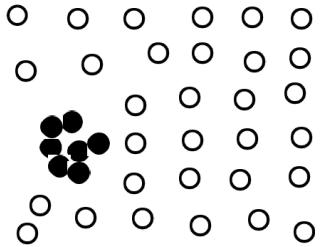
- Three kinds: *global, contextual* and *collective* outliers
- **Global outlier** (or point anomaly)
  - Object is O_g if it significantly deviates from the rest of the data set
  - Ex. Intrusion detection in computer networks
  - Issue: Find an appropriate measurement of deviation



- **Contextual outlier** (or *conditional outlier*)
  - Object is O_c if it deviates significantly based on a selected context
  - Ex. 35° C in Gold Coast: outlier? (depending on summer or winter?)
  - Attributes of data objects should be divided into two groups
    - Contextual attributes: defines the context, e.g., time & location
    - Behavioral attributes: characteristics of the object, used in outlier evaluation, e.g., temperature
  - Can be viewed as a generalization of *local outliers*—whose density significantly deviates from its local area

Types of Outliers (II)

- **Collective Outliers**
  - A subset of data objects *collectively* deviate significantly from the whole data set, even if the individual data objects may not be outliers
  - Applications: E.g., *intrusion detection*:
    - When a number of computers keep sending denial-of-service packages to each other
  - In the Detection of collective outliers
    - Consider not only behavior of individual objects, but also that of groups of objects
    - Need to have the background knowledge on the relationship among data objects, such as a distance or similarity measure on objects.
- A data set may have multiple types of outlier
- One object may belong to more than one type of outlier



Collective Outlier

Challenges of Outlier Detection

- Modeling normal objects and outliers properly
  - Hard to list all possible normal behaviors in an application
  - The border between normal and outlier objects is often a grey area
- Application-specific outlier detection
  - Choice of distance measure among objects and the model of relationship among objects are often application-dependent
  - E.g., clinic data: a small deviation could be an outlier; while in marketing analysis, larger fluctuations
- Handling noise in outlier detection
  - Noise may distort the normal objects and blur the distinction between normal objects and outliers. It may help **hide outliers** and reduce the effectiveness of outlier detection
- Understandability
  - Understand why these are outliers: Justification of the detection
  - Specify the degree of an outlier: the unlikelihood of the object being generated by a normal mechanism

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Outlier Detection I: Supervised Methods

- Two ways to categorize outlier detection methods:
  - Based on whether user-*labeled* examples of outliers can be obtained:
    - Supervised, semi-supervised vs. unsupervised methods
  - Based on *assumptions about normal data and outliers*:
    - Statistical, proximity-based, and clustering-based methods
- **Outlier Detection I: Supervised Methods**
  - Modeling outlier detection is a classification problem
    - Samples examined by domain experts can be used for training & testing
  - Methods for Learning a classifier for outlier detection is effective
    - Model normal objects & report those not matching the model as outliers, or
    - Model outliers and treat those not matching the model as normal
  - Challenges
    - Imbalanced classes, i.e., outliers are rare: Boost the outlier class and make up some artificial outliers
    - Catch as many outliers as possible, i.e., recall is more important than accuracy (i.e., not mislabeling normal objects as outliers)
      - ❑ Accuracy *acc* is the number of correctly classified examples (including positive and negative) divided by the total number of all data samples.
      - ❑ Recall *r* is the number of correctly classified positive examples divided by the total number of actual positive examples in the test set.

Outlier Detection II: Unsupervised Methods

- Assume the normal objects are somewhat "clustered" into multiple groups, each having some distinct features
- An outlier is expected to be far away from any groups of normal objects
- Weakness: Cannot detect collective outlier effectively
  - Normal objects may not share any strong patterns, but the collective outliers may share high similarity in a small area
- Many clustering methods can be adapted for unsupervised methods
  - Find clusters, then outliers: not belonging to any cluster
    - Problem 1: Hard to distinguish noise from outliers
    - Problem 2: Costly since first clustering: but far less outliers than normal objects
      - Newer methods: tackle outliers directly

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Statistical Approaches

- Statistical approaches assume that the objects in a dataset are generated by a generative model
  - *Given an observable variable X and a target variable Y, a generative model is a model of the conditional probability of the observation X, given a target y, symbolically, P(X|Y=y)*
- **Idea**: learn a generative model fitting the given data set, and then identify the objects in low probability regions of the model as outliers
- Methods are divided into two categories: *parametric* vs. *non-parametric*
- Parametric method
  - Assumes that the normal data is generated by a parametric distribution with parameter θ
- Non-parametric method
  - Not assume an a-priori statistical model and determine the model from the input data

Parametric Methods: Normal Distribution

- Univariate data: A data set involving only one attribute or variable
- Often assume that data are generated from a normal distribution, learn the parameters from the input data, and identify the points with low probability as outliers
- Ex: Avg. temp.: {24.0, 28.9, 28.9, 29.0, 29.1, 29.1, 29.2, 29.2, 29.3, 29.4}
  - Use the maximum likelihood method to estimate μ and σ

$$\ln \mathcal{L}(\mu, \sigma^2) = \sum_{i=1}^{n} \ln f(x_i|(\mu, \sigma^2)) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2 \qquad \varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

  - Taking derivatives with respect to μ and σ^2, we derive the following maximum likelihood estimates

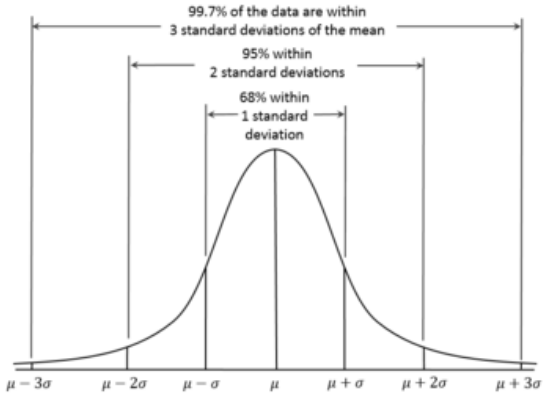
$$\hat{\mu} = \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

  - For the above data with n = 10, we have

$$\hat{\mu} = 28.61 \qquad \hat{\sigma} = \sqrt{2.29} = 1.51$$

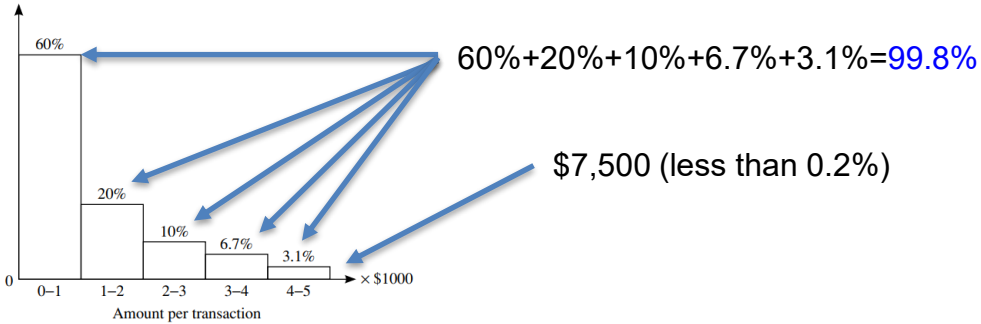
  - Then (24 – 28.61) /1.51 = – 3.04 < –3, 24 is an outlier since

$$\mu \pm 3\sigma \text{ region contains } 99.7\% \text{ data}$$



Non-Parametric Methods: Detection Using Histogram

- The model of normal data is learned from the input data without any *a-priori* structure.
- Often makes fewer assumptions about the data, and thus can be applicable in more scenarios
- Outlier detection using histogram:
  - Figure shows the histogram of purchase amounts in transactions
  - A transaction in the amount of $7,500 is an outlier, since only 0.2% transactions have an amount higher than $7,500



- Problem: Hard to choose an appropriate bin size for histogram
  - Too small bin size → normal objects in empty/rare bins, false positive (report an outlier for a normal observation)
  - Too big bin size → outliers in some frequent bins, false negative (missing an actual outlier, as more data are included in a wider bin)
- Solution: Adopt kernel density estimation to estimate the probability density distribution of the data. If the estimated density function is high, the object is likely normal. Otherwise, it is likely an outlier.

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Proximity-Based Approaches

- Intuition: Objects that are far away from the others are outliers
- Assumption of proximity-based approach: The proximity of an outlier deviates significantly from that of most of the others in the data set
- **Two types of proximity-based** outlier detection methods
  - Distance-based outlier detection: An object o is an outlier if its neighborhood does not have enough other points
  - Density-based outlier detection: An object o is an outlier if its density is relatively much lower than that of its neighbors

Distance-Based Outlier Detection

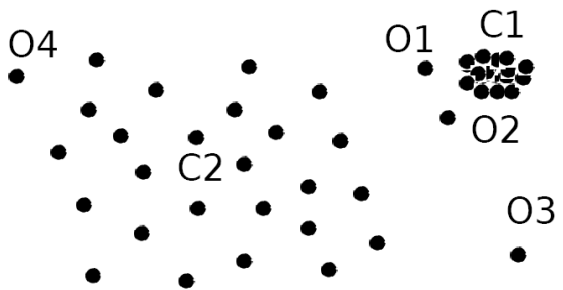
- For each object o, examine the # of other objects in the *r*-neighborhood of o, where *r* is a user-specified **distance threshold**
- An object o is an outlier if most (taking π as a **fraction threshold**) of the objects in D are far away from o, i.e., not in the r-neighborhood of o

$$\frac{\|\{o' | dist(o, o') \leq r\}\|}{\|D\|} \leq \pi$$

- Equivalently, one can check the distance between *o* and its *k*-th nearest neighbor o_k, where *o* is an outlier if dist(*o*, o_k) > r $\quad k = \lceil \pi \|D\| \rceil$
- Efficient computation: Nested loop algorithm
  - For any object o_i, calculate its distance from other objects, and count the # of other objects in the r-neighborhood.
  - If π·n other objects are within r distance, terminate the inner loop
  - Otherwise, o_i is a DB(r, π) outlier
- Efficiency: Actually CPU time is not $O(n^2)$ but linear to the data set size since for most non-outlier objects, the inner loop terminates early

Density-Based Outlier Detection

- Local outliers: Outliers comparing to their local neighborhoods, instead of the global data distribution
- In Figure below, o_1 and o_2 are local outliers to C_1, o_3 is a global outlier, but o_4 is not an outlier. However, distance-based clustering cannot find o_1 and o_2 are outlier (e.g., comparing with O_4).
- Intuition (density-based outlier detection): The density around an outlier object is significantly different from the density around its neighbors
- Method: Use the *relative density* of an object against its neighbors as the indicator of the degree of the object being outliers
- *k-distance* of an object o, $dist_k(o)$: distance between o and its k-th NN
- *k-distance neighborhood* of o, $N_k(o) = \{o' | o' \text{ in } D, dist(o, o') \leq dist_k(o)\}$
  - $N_k(o)$ could be bigger than k since multiple objects may have identical distance to o



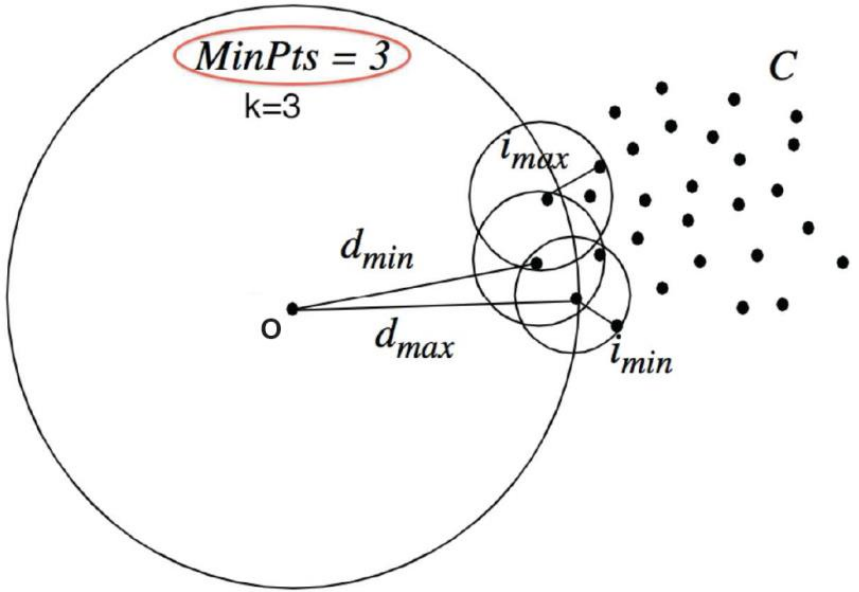
Density-Based Outlier Detection

- Reachability distance from *o'* to *o*: $reachdist_k(o \leftarrow o') = \max\{dist_k(o), dist(o, o')\}$
  - where k is a user-specified parameter

- Local reachability density of *o*:

$$lrd_k(\boldsymbol{o}) = \frac{\|N_k(\boldsymbol{o})\|}{\sum_{\boldsymbol{o'} \in N_k(\boldsymbol{o})} reachdist_k(\boldsymbol{o'} \leftarrow \boldsymbol{o})}.$$

- This captures a local outlier whose local density is relatively low comparing to the local densities of its kNN

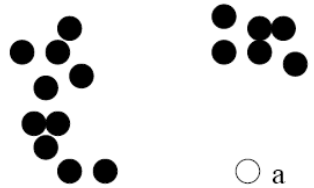


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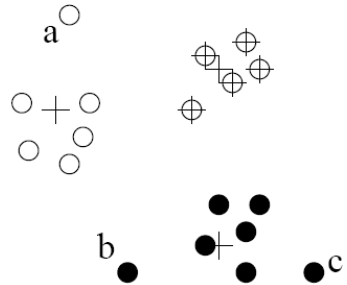
Clustering-Based Outlier Detection (1 & 2)

- An object is an outlier if (1) it does not belong to any cluster, (2) there is a large distance between the object and its closest cluster , or (3) it belongs to a small or sparse cluster



- Case I: Not belong to any cluster
  - Using a density-based clustering method such as DBSCAN

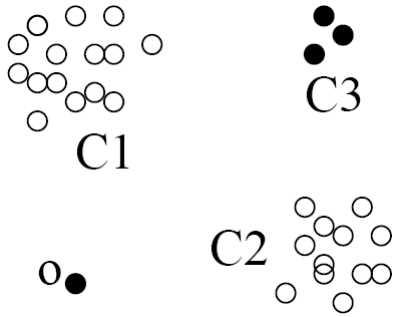
- Case 2: Far from its closest cluster
  - Using k-means, partition data points into clusters
  - For each object o, assign an outlier score based on its distance from its closest center
    - If dist(o, c_o)/avg_dist(c_o) is large, likely an outlier



where avg_dist(c_o) is the average distance between c_o and the objects assigned to o

Clustering-Based Outlier Detection (3)

- *Case 3: outliers may be similar and form a small cluster.*
- *FindCBLOF:* Detect outliers in small clusters
  - Find clusters, and sort them in decreasing size
  - To each data point, assign a *cluster-based local outlier factor* (CBLOF)
    - If object p belongs to a large cluster, CBLOF = cluster_size * similarity between p and cluster
    - If p belongs to a small one, CBLOF = cluster size * similarity between p and the closest large cluster
    - Ex. In the figure, o is outlier since its closest large cluster is C_1, but the similarity between o and C_1 is small. For any point in C_3, its closest large cluster is C_2 but its similarity from C_2 is low, plus $|C_3| = 3$ is small



Set a threshold to define large cluster: any cluster size greater than α of the data is considered as a large cluster

Clustering-Based Method: Strength and Weakness

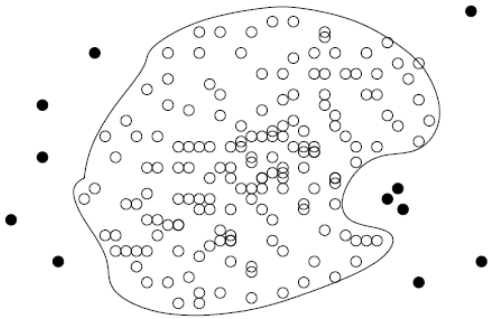
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- **Strength**
  - Detect outliers without requiring any labeled data
  - Work for many types of data
  - Clusters can be regarded as summaries of the data
  - Once the cluster are obtained, need only compare any object against the clusters to determine whether it is an outlier (fast)
- **Weakness**
  - Effectiveness depends highly on the clustering method used—they may not be optimized for outlier detection
  - High computational cost: Need to first find clusters

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Classification-Based Method I: One-Class Model



- **Idea**: Train a classification model that can distinguish "normal" data from outliers
- **A brute-force approach**: Consider a training set that contains samples labeled as "normal" and others labeled as "outlier"
  - But the training set is typically heavily biased: # of "normal" samples likely far exceeds # of outlier samples
  - Cannot detect unseen anomaly

- **One-class model**: A classifier is built to describe only the normal class.
  - Learn the decision boundary of the normal class using classification methods such as SVM
  - Any samples that do not belong to the normal class (not within the decision boundary) are declared as outliers
  - Pros: can detect new outliers that may not appear close to any outlier objects in the training set
  - Extension: Normal objects may belong to multiple classes

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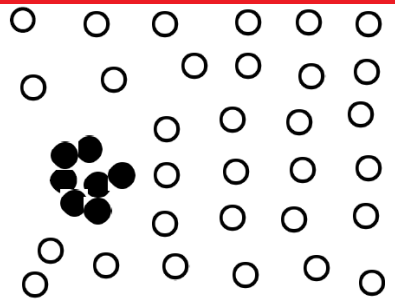
Mining Contextual Outliers

- If the contexts can be clearly identified, transform it to **conventional outlier detection**
  - Identify the context of the object using the contextual attributes
  - Calculate the outlier score for the object in the context using a conventional outlier detection method

- Ex. Detect outlier customers in the context of customer groups
  - Contextual attributes: *age group, postal code*
  - Behavioral attributes: *# of trans/yr*, *annual total trans. amount*

- Steps:
  (1) *locate c's context*,
  (2) *compare c with the other customers in the same group*, and
  (3) *use a conventional outlier detection method.*

Mining Collective Outliers



- Collective outliers are formed if objects as a group deviate significantly from the entire data
- Need to examine the structure of the dataset, i.e, the relationships between multiple data objects
- Each of these structures is inherent to its respective type of data
  - For temporal data (such as time series and sequences), we explore the structures formed by time, which occur in segments of the time series or subsequences
  - For spatial data, explore local areas
  - For graph and network data, we explore subgraphs
- Collective outlier detection methods:
  - Reduce the problem to conventional outlier detection
    - Identify *structure units*, treat each structure unit (e.g., subsequence, time series segment, local area, or subgraph) as a data object, and extract features
    - Then outlier detection **on the set of "structured objects"** can be done using the extracted features

Summary

- Types of outliers
  - global, contextual & collective outliers
- Outlier detection
  - Supervised & unsupervised
- Statistical (or model-based) approaches
- Proximity-based approaches
- Clustering-based approaches
- Classification approaches
- Mining contextual and collective outliers

Have a question?

